Revision History

Date	Version	Description
July 19, 2019	0.10	Initial version created.
August 6, 2019	0.20	Updated various sections.
February 14, 2020	0.30	Updated bullet 3.
June 13, 2020	0.40	Updated various sections.
March 29, 2021	0.50	Updated bullet 3.
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Pin Assignment and Pinmux Programming Considerations

ATM2/3 Series has up to 34 pins labelled P<*N*> where <*N*> is an integer. Throughout the remainder of this document, this collection of pins will be referred to as the P pins.

An individual P pin can be one of multiple functions based on which options are allowed by the pinmux logic and on how the pinmux pin selection registers are programmed. Using the Atmosic Pinmux Tool in the SDK is recommended for selecting digital IO functionality.

Some P pins are already assigned to specific functions by the ATM2/3. Before selecting P pins for application functionality, a user should take these items into consideration:

1. If the application is executing from external flash, a QSPI interface must be present and it cannot be time shared. That locks 6 P pins (QSPI_CS, QSPI_CLK, and QSPI_DIO[3:0]).
2. The SDK uses the SWD interface to download code and for debug access. For the ATMx2x1, asserting TMC high enables the SWD interface by making P0 the JTAG select control line, P1 the SWD interface clock (SWD_CLK), and P2 the SWD interface bidirectional data (SWD_IO). P0 through P2 can still be used for a user specified function but that function would not be available when the SWD interface is enabled.
3. The ATMx2xx-x0x ROM code uses P3/P4/P5/P6/P7/P8/P9 to keep track of various internal states. As a result, those pins cannot be used as GPIO outputs, but the ATMx2xx-x0x GPIO driver in the SDK supports configuring them as open-drain outputs with internal pull-ups. The driver uses Bluetooth coexistence functionality for P3 through P6 and PWM functionality for P7 through P9. A user application can also configure the pins for non-GPIO functionality or as GPIO inputs, but they will still be driven low by the ROM code until overridden. For the ATMx2xx-x1x, P3/P4/P5/P6/P7/P8/P9 can be used as GPIO outputs without any limitations. Refer to the ATMx2xx Chip Revisions Application Note for information regarding chip revisions.
4. The ROM code configures P33 to be UART1 TX so that debug and run time information can be displayed. This P pin can still be used for a user specified function, but that function would not be available when the debug UART is enabled.
5. P32 is a strap option for benign boot that, if high, instructs the M0 CPU to stop the boot process and go into an idle state. This P pin can still be used for a user specified function, but that function would need to avoid pulling or driving the pin high whenever the CPU boots.

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A Case Study - ATMx221 HID_keyboard SDK Example

This case study examines the ATMx221 pin selection for the HID_keyboard SDK example. The target keyboard matrix is 13x10.

The application requires an external flash, so P14/P15/P16/P17/P18/P19 need to be assigned to QSPI.

Next, P0/P1/P2 are assigned to SWD so that the SDK can download code and enable debug access.This decision will only be reconsidered if there are not enough remaining P pins.

Next, P32 is assigned to benign boot while P33 is assigned to debug UART for full debug control and access. This decision will only be reconsidered if there are not enough remaining P pins.

At this point, 23 unassigned pins remain, which is exactly the number needed for the 13x10 keyboard matrix. Because P3/P4/P5/P6/P7/P8/P9 are being assigned to non-GPIO functionality, there is no concern about their use by the ROM code.

The complete pin assignment is shown in the following table:

P Pin	Assigned Function	P Pin	Assigned Function
P0	JTAG select	P17	QSPI
P1	SWD clock	P18	QSPI
P2	SWD data	P19	QSPI
P3	Keyboard matrix	P20	Keyboard matrix
P4	Keyboard matrix	P21	Keyboard matrix
P5	Keyboard matrix	P22	Keyboard matrix
P6	Keyboard matrix	P23	Keyboard matrix
P7	Keyboard matrix	P24	Keyboard matrix
P8	Keyboard matrix	P25	Keyboard matrix
P9	Keyboard matrix	P26	Keyboard matrix
P10	Keyboard matrix	P27	Keyboard matrix
P11	Keyboard matrix	P28	Keyboard matrix
P12	Keyboard matrix	P29	Keyboard matrix

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P13	Keyboard matrix		P30	Keyboard matrix
P14	QSPI		P31	Keyboard matrix
P15	QSPI		P32	Benign boot
P16	QSPI		P33	Debug UART

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